

Machine_Learning_Engineer — Step 6

Deploy ML Models

Detailed Learning Notes • CareerCompass

1. Learning Objective

Turn a trained model into a reproducible inference service or application while handling preprocessing, versioning, monitoring and operational risks.

2. Core Concepts — Detailed Breakdown

Training vs inference

Training produces model parameters from data; inference applies a fixed artifact to new inputs. The exact preprocessing used during training must be applied consistently at inference.

Model serialization

Save the model and its preprocessing pipeline together where possible. Record library versions and metadata so the artifact can be reproduced and loaded safely.

Serving patterns

A model can run inside a batch job, a web API, a background worker or an embedded application. Choose based on latency, throughput, update frequency and operational complexity.

Input validation

APIs should validate types, ranges, required fields and categorical values before inference. Bad input should produce a controlled error rather than silently corrupting predictions.

Monitoring

Track latency, errors, input distribution changes, output distribution and model quality when labels become available. Data drift does not automatically mean model failure, but it is a signal to investigate.

Security and lifecycle

Protect model endpoints, avoid unsafe deserialization, control dependencies and define how models are replaced or rolled back.

3. Recommended Learning Workflow

- Package preprocessing and model together.
- Create a deterministic inference function.
- Expose a small API or batch interface.
- Add validation and structured logging.
- Test normal, invalid and edge-case inputs.
- Document deployment and rollback steps.

5. Hands-on Project / Practice

Deploy a classification model behind a small REST API. The endpoint should accept JSON, validate fields, run the saved preprocessing/model pipeline, return a prediction plus model version, log latency and reject malformed requests. Add a README explaining how to run and test it locally.

6. Common Mistakes & Pitfalls

- Saving only the estimator but not preprocessing.
- Loading untrusted model files.
- Training and serving with different feature order or encodings.
- No input validation or versioning.
- Assuming a successful HTTP response means the model is healthy.

7. Completion Checklist

- I can separate training from inference.
- I can save/load a complete preprocessing pipeline.
- I can expose a prediction API or batch job.
- I can validate inputs and log inference.
- I can describe basic drift and monitoring signals.

8. Interview / Assessment Questions

- What is the difference between training and inference?
- Why must preprocessing be identical at inference time?
- What should a model API validate?
- What metrics should be monitored in production?
- What is model drift?

9. Recommended References

Scikit-learn User Guide: https://scikit-learn.org/stable/user_guide.html

TensorFlow/Keras: <https://www.tensorflow.org/guide/keras>

Machine Learning in 2024 — freeCodeCamp: <https://www.youtube.com/watch?v=bmmQA8A-yUA>

Note: These notes are designed as a structured learning guide. Use the references for deeper, current documentation and hands-on practice.